PROXMOX PRODUCTION VALIDATION ANALYZED AUDIT REPORT

Cluster, Ceph, ZFS, Storage and High Availability Control Review

Document owner	Nishu Saini
Function	CISAI IT Support / Infrastructure
Source evidence	proxmox_validation_prox24_20260905_093952.log
Cluster	cisai-pve
Validation host	prox24
Validation timestamp	2026-09-05 09:39:52 +05:30
Execution mode	READ-ONLY
Classification	Internal - Production Audit Evidence
Document version	1.0

OVERALL RESULT: PASS WITH WARNINGS

The source report recorded 42 PASS, 2 WARN and 0 FAIL. The warnings identify live migration and disruptive node-failure simulation as not executed in the read-only run.

This report analyzes and summarizes the supplied system-generated log. The original log and its recorded SHA-256 value remain the detailed audit evidence.

1. Executive Assessment

The validation evidence records a quorate three-node Proxmox cluster with active core cluster services, HEALTH_OK Ceph storage, an ONLINE local RAIDZ2 pool on prox24, active configured storage, and active HA management services. No failed check was recorded. The two warnings are coverage gaps caused by the read-only execution mode, not recorded infrastructure faults.

Control domain	Recorded evidence	Assessment
Platform and host	Proxmox VE 9.2.0; pve-manager 9.2.11; kernel 7.0.14-14-pve	PASS
Time and root filesystem	NTP synchronized; root filesystem 10% used	PASS
Cluster	3 nodes; Quorate: Yes; pve-cluster and corosync active	PASS
Ceph	HEALTH_OK; 3 MONs; active MGR with 2 standbys; 12 OSDs up/in	PASS
Placement groups	33 active+clean	PASS
ZFS	local-zfs-prox24 ONLINE; RAIDZ2; 7 members; no known data errors	PASS
Storage availability	ceph, local, local-lvm and local-zfs-prox24 active	PASS
HA services	HA manager quorum OK; pve-ha-lrm and pve-ha-crm active	PASS
Migration and failure testing	Not executed in read-only mode	OPEN TESTS

Production interpretation

The evidence supports a successful read-only production health validation. It does not provide execution proof for live VM migration or controlled HA recovery, because the source log explicitly states that those tests were not performed.

2. Scope, Method and Limitations

Area	Included evidence
Pre-flight	Required commands, product versions, time synchronization and root capacity
Cluster	Membership, quorum, pve-cluster, corosync and recent Corosync warning/error check
Ceph	Summary, health detail, MON quorum, MGR status, OSD topology, utilization, latency and capacity
ZFS	Pool existence, status, RAIDZ2 members, capacity, datasets and data-error state
Storage	Proxmox storage status and active/disabled node-local entries
VM and HA	VM inventory, HA manager state and HA service status

Limitation: The run was READ-ONLY. Performance load testing, VM live migration and disruptive node-failure behavior were not part of this execution.

3. Platform and Cluster Analysis

Check	Recorded value	Assessment
Validation script	Proxmox Production Validation v1.1	RECORDED
Host	prox24	RECORDED
Proxmox VE	9.2.0	RECORDED
pve-manager	9.2.11/f6997e698c7933ea	RECORDED
Kernel	7.0.14-14-pve	RECORDED
Ceph package	20.2.2-pve1	RECORDED
ZFS utilities	2.4.4-pve1	RECORDED
Time synchronization	Timezone Asia/Kolkata; NTPSynchronized=yes	PASS
Root filesystem	94G size; 8.1G used; 82G available; 10% use	PASS

Cluster control	Recorded evidence	Assessment
Cluster name	cisai-pve	RECORDED
Transport and authentication	knet; secure auth on	PASS
Membership	prox24, prox22 and prox23	PASS
Voting	Expected 3; total 3; quorum 2	PASS
Quorate state	Yes	PASS
Cluster services	pve-cluster active; corosync active	PASS
Recent Corosync warnings/errors	No entries	PASS

4. Ceph Storage Analysis

Control	Recorded evidence	Assessment
Overall health	HEALTH_OK	PASS
MON quorum	3 daemons: prox24, prox22, prox23; leader prox24	PASS
MGR	prox23 active; 2 standbys	PASS
OSDs	12 total; 12 up; 12 in	PASS
OSD distribution	4 OSDs on each of prox22, prox23 and prox24	PASS
Placement groups	33 active+clean	PASS
Raw capacity	218 TiB total; 588 MiB used in the captured output	RECORDED
Pools	.mgr and ceph pools listed	RECORDED
Latency indicator	All listed commit/apply values shown as 0 ms	PASS
Final Ceph validation	HEALTH_OK; all OSDs up	PASS

Analysis note

The evidence shows balanced OSD count by host and healthy placement-group state at the validation timestamp. The read-only validation does not constitute a sustained performance benchmark.

5. ZFS and Proxmox Storage Analysis

Control	Recorded evidence	Assessment
Pool	local-zfs-prox24	RECORDED
Pool state	ONLINE	PASS
Layout	raidz2-0	PASS
Members	7 WWN-identified members, all ONLINE	PASS
Error counters	READ 0; WRITE 0; CKSUM 0	PASS
Known data errors	No known data errors	PASS
Capacity	127T size; 1.79M allocated; 127T free; 0% capacity	PASS
Fragmentation	0%	PASS
Datasets	local-zfs-prox24 and local-zfs-prox24/pve listed	PASS

Storage	Type	Status on prox24	Interpretation
ceph	rbd	active	PASS
local	dir	active	PASS
local-lvm	lvmthin	active	PASS
local-zfs-prox24	zfspool	active	PASS
local-zfs-prox22	zfspool	disabled	Informational: node-restricted entry
local-zfs-prox23	zfspool	disabled	Informational: node-restricted entry

The source log classifies the prox22 and prox23 ZFS entries as informational node-restricted storage on prox24. No corrective action is established by this evidence.

6. HA and Resilience Coverage

Control	Recorded evidence	Assessment
HA manager	quorum OK	PASS
CRM master	prox22, idle	RECORDED
Fencing state	standby; CRM watchdog standby	RECORDED
LRM state	prox22, prox23 and prox24 idle with watchdog standby	PASS
pve-ha-lrm	active	PASS
pve-ha-crm	active	PASS
Live migration	Not executed	OPEN
Controlled node-failure simulation	Not performed	OPEN

7. Findings, Risks and Recommended Closure Evidence

ID	Finding	Evidence-based impact	Closure evidence
OBS-01	Live migration was not executed.	The report does not demonstrate movement of a running VM between nodes.	Retain a controlled test record identifying the test VM, source, target, command outcome and post-migration guest availability.
OBS-02	Disruptive node-failure simulation was not performed.	The report does not demonstrate workload recovery after a node failure.	Retain an approved HA test record showing pre-checks, fencing/watchdog state, recovery outcome and post-test health.
INFO-01	Peer node ZFS stores are disabled on prox24.	The source script labels these as node-restricted storage entries.	No closure needed unless intended node restrictions differ from the approved storage design.

Risk statement

No operational failure is recorded in the supplied evidence. Residual audit risk is limited to untested resilience behaviors explicitly excluded from the read-only run.

8. Final Audit Disposition

PASS WITH WARNINGS

42 passed checks, 2 warnings and 0 failures. The infrastructure health controls tested in read-only mode passed. Live migration and controlled HA failure/recovery remain open for separate evidence.

Role	Name	Signature	Date
Prepared by	Nishu Saini		
Technical reviewer			
Change owner / approver			

Appendix A - Evidence Traceability

Report claim	Source section
Versions, NTP and root filesystem	1. PRE-FLIGHT
Membership, quorum and core services	2. PROXMOX CLUSTER HEALTH AND QUORUM
Ceph health, MON, MGR, OSD, PG and capacity	3. CEPH HEALTH, COMPONENTS, AND PERFORMANCE INDICATORS
ZFS pool, RAIDZ2 layout and errors	4. ZFS STORAGE VALIDATION
Active and node-restricted storage	5. SHARED STORAGE AVAILABILITY
VM inventory and live-migration warning	6. VM INVENTORY AND MIGRATION READINESS; 7. LIVE MIGRATION TEST
HA services and failover warning	8. HA CONFIGURATION AND CONTROLLED RECOVERY TEST
42 PASS, 2 WARN, 0 FAIL and SHA-256	9. FINAL SUMMARY

Appendix B - Source Evidence Integrity

Source file: proxmox_validation_prox24_20260905_093952.log

Recorded SHA-256: f6ec059bbc2ed1078dcf61266d6545969749596a6e500c13e1c536b743df3dd7

Retain this PDF together with the original source log. Preserve the original filename, timestamp and recorded checksum. Store later validation runs as separate evidence sets rather than overwriting this evidence.